ENT-164 · Intro to Making

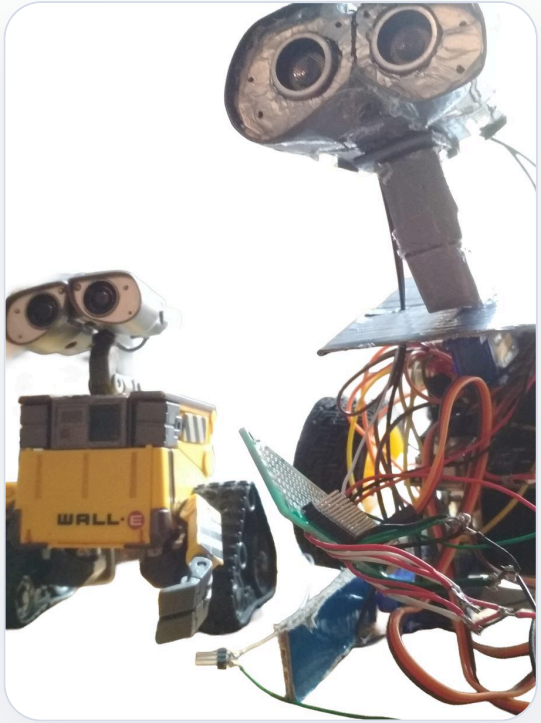
Tufts University | Derby Entrepreneurship Center at Tufts

CLASS 6 · WEEK 6 · FALL 2026

Going Wireless

Your ESP32 can serve a web page your phone opens. No app required. Today we get your robot on WiFi, build it up step by step, and drive it from your phone's browser.

Thu, Oct 15 | JCC 301 / Nolop | No programming required | Bring your ESP32



Thomas Van de Velde · Instructor · **Alexander Beattie** · TA · **Patrice Pinardo** · TA

Tufts University · Gordon Institute

PHYSICAL MAKING · REFLECTIONS

Catching the surprise

What your reflections show, and how to make them sharper.

What you noticed

Three patterns from your Physical Making reflections.

`7 of 10`

Parts didn't fit

Laser kerf, wood thickness, finger joints that wouldn't align, interior vs exterior dimensions.

`Jams`

Things didn't behave

Balls stuck in funnels. A stand that fell over. Wheels glued before anyone knew how they turned.

`Plan`

The usual lesson

Most reflections ended with "measure and plan more next time."

Four ways to sharpen a reflection

Each one rewrites a moment so it teaches you something you can reuse.

01

Name what you expected

Instead of “The notches were too loose.”

Try “I assumed the cut matched my drawing. The laser removed extra wood.”

So that you know which belief to update next time.

02

Treat the fix as a test

Instead of “We filled the gaps with hot glue.”

Try “I cut one notch 0.2 mm wider to check the fit first.”

So that the fix teaches you a rule you can reuse.

03

Reframe, don’t just repair

Instead of “We made the funnel opening smaller.”

Try “The balls jam at the opening, so I redesigned around how they roll.”

So that the next design starts from how things behave.

04

Look past “plan more”

Instead of “Next time I’ll plan and measure more.”

Try “Next time I’ll build a cheap version first.”

So that you find surprises early, while they’re cheap.

PUT IT TO WORK

The goal isn't fewer surprises. It's cheaper ones.

Cut one first

One joint or test piece before the full sheet.

Write the guess

Before you build, note what you expect to happen.

Assemble virtually

Put the parts together in Onshape before cutting.

Seven weeks from sketch to race day

Today is Week 6. Going Wireless. Here's how the robot build comes together.

Week 2

Hand Making and Robot Challenge Kickoff

Sketch concept variations; build a foam core mockup.

Week 3

Laser Cutting

Create a laser-ready 2D design; cut structural parts.

Week 4

3D Printing

Model 3D parts; 3D print functional parts.

Week 5

Electronics

Wire motors, sensors, and controls; program with MicroPython + AI.

Week 6

Going Wireless

Light an RGB LED from your phone; drive the robot over its own WiFi.

Week 7

Final Robot Assembly and Testing

Final integration and test runs.

Week 8

Robo Challenge and Team Retrospective

Race on the course, then close out the first module.

TODAY **Week 6 · Going Wireless**. Light an RGB LED from your phone, wire up the robot, and drive it over its own WiFi hotspot.

Before we go wireless

You should have all three from last class. Tell us now if something is missing.

01

Electronics kit

You've received your electronics kit. The parts you'll wire up today.

02

opencode + MicroPython

opencode is set up and MicroPython is deployed to your ESP32. It runs your code on the board.

03

AI-assisted electronics

You can already build basic electronics projects with the help of AI.

From your phone to the board

Connect, then build. The two things we do in class today.

01

Connect to your microcontroller

Reach your device from a phone and from a web browser. No native app needed.

02

Drive your robot

Put it all together: a control page on your phone that drives the robot you built.

Direct — the board's own hotspot

The board becomes the network: your phone joins it, and nothing else is needed.

01

The board is the network

The ESP32 creates its own WiFi hotspot. Your phone joins it like any network — no router involved.

02

Join it, open the page

The board serves the page at its own address — like `192.168.4.1`. Type it in and the controls are there.

03

Simple, but offline

While your phone is on the board's network it is off the internet — normal, and fine for class.

Shared — through the internet

The board joins a network you already have, and you reach it over that network.

01

One network, two devices

The board joins a WiFi — home, or a shared one — and gets an address on it, like any computer.

02

Reach it from anywhere on the network

Any device on that network can open the board's page — and your phone keeps its internet while you control it.

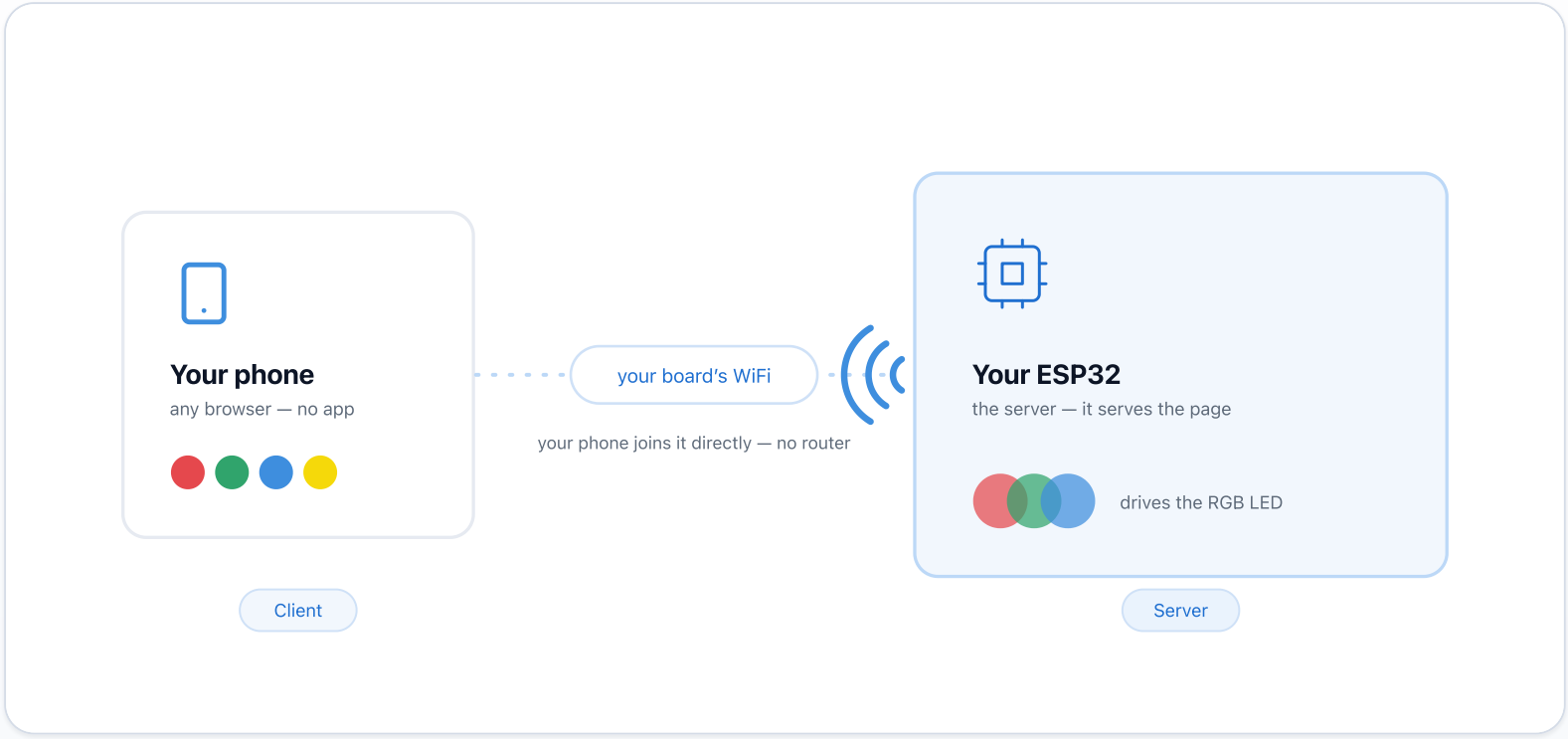
03

It also gets the internet

On a shared network the board can reach online services — and it obeys the network's rules, like its password.

Client and server

Your phone asks for the page; the board serves it — over the board's own WiFi. No router, no cloud.



Client — your phone asks · **Server** — the board answers · **No cloud** — the board is the whole network.

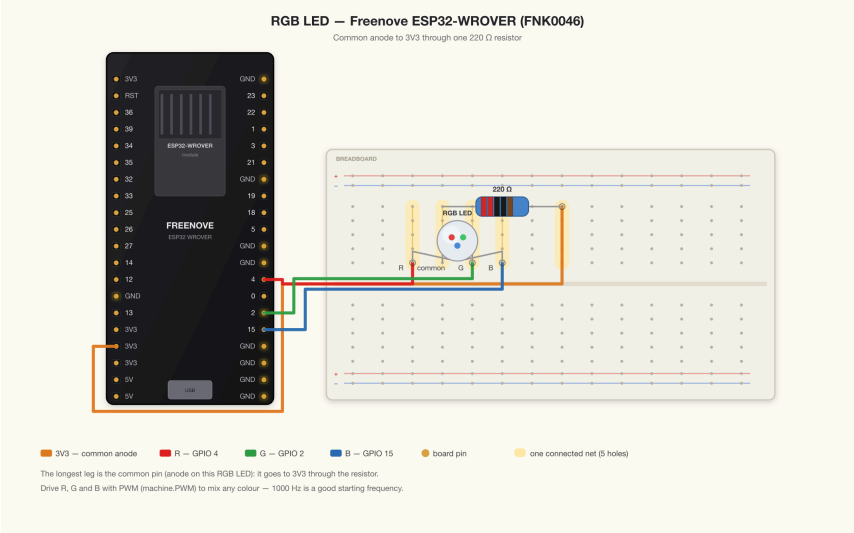
Change a color from your phone

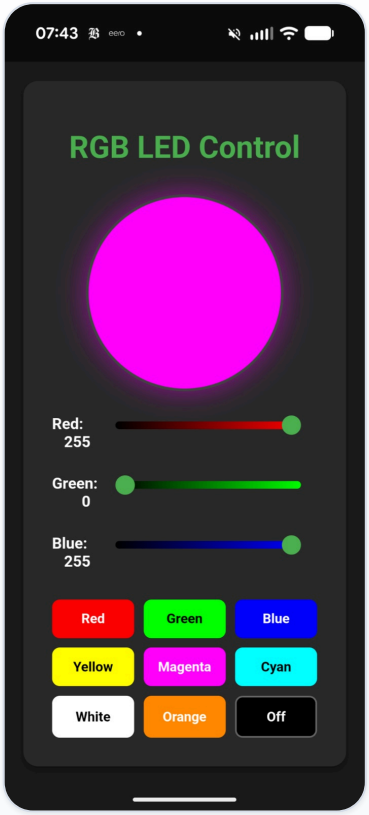
Before the robot, prove the whole loop on one LED: wire it, ask for a page, open it on your phone.

THE PROMPT

“I’ve connected an ESP32. I want to use MicroPython to control an RGB LED from my phone over a wireless hotspot.”

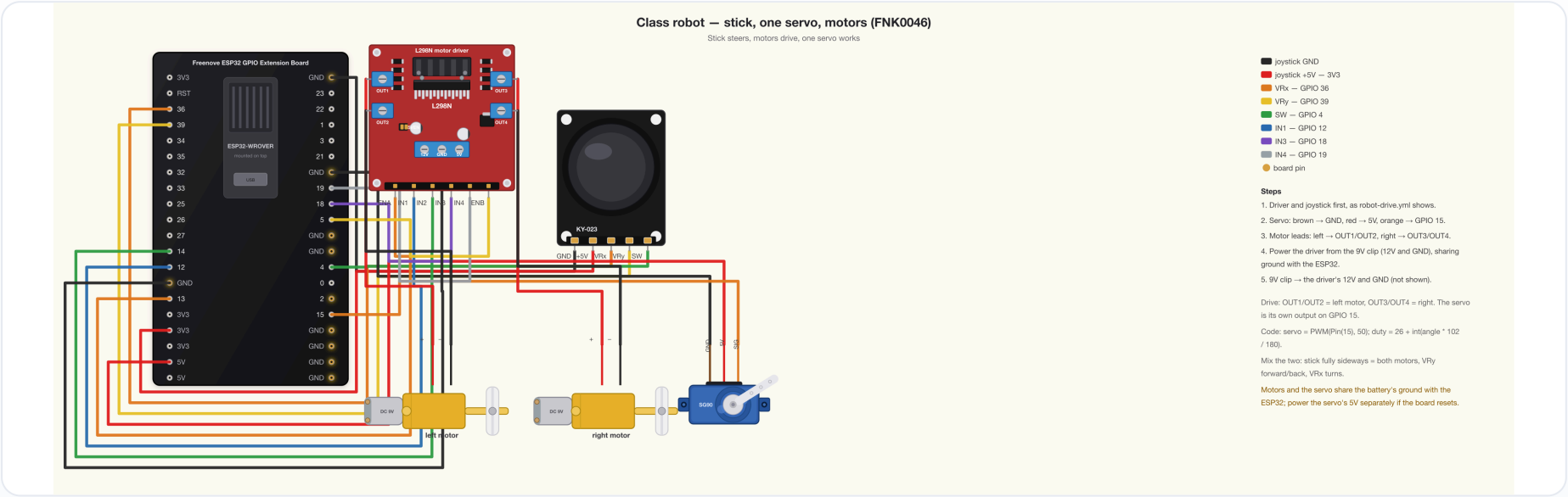
Join the board’s WiFi, open the address it prints, and pick a color.





At Nolop: Wiring the drive train

The L298N sits between your ESP32 and the motors — the same wiring your robot already has.



Wheels **off the ground** while you test. The motors spin the moment the code runs.

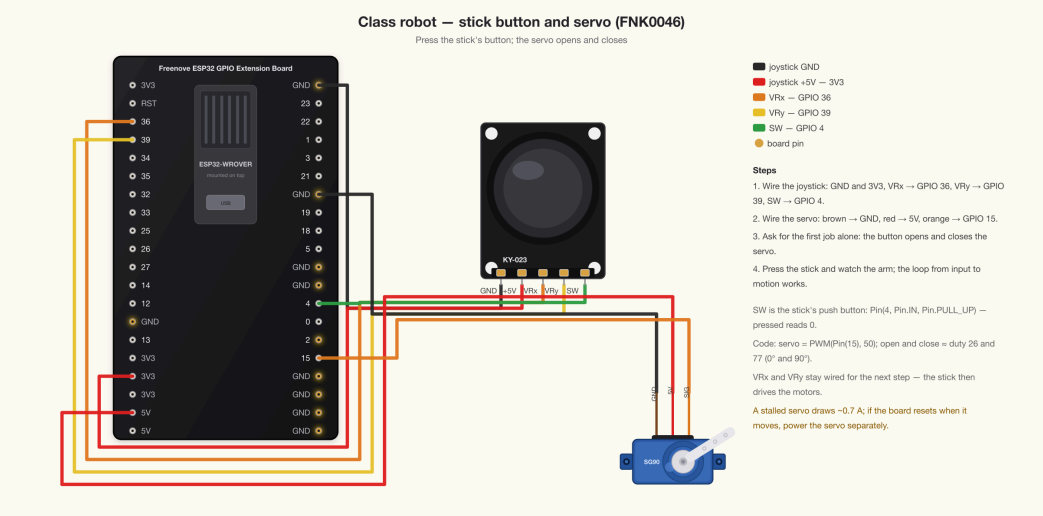
Start with one button and one servo

The smallest piece of the robot: press, and the arm moves. Prove the loop before adding motors.

THE PROMPT

“I want to use my joystick’s push button to open and close the servo.”

One small ask, one test. This is the loop for every step after it.



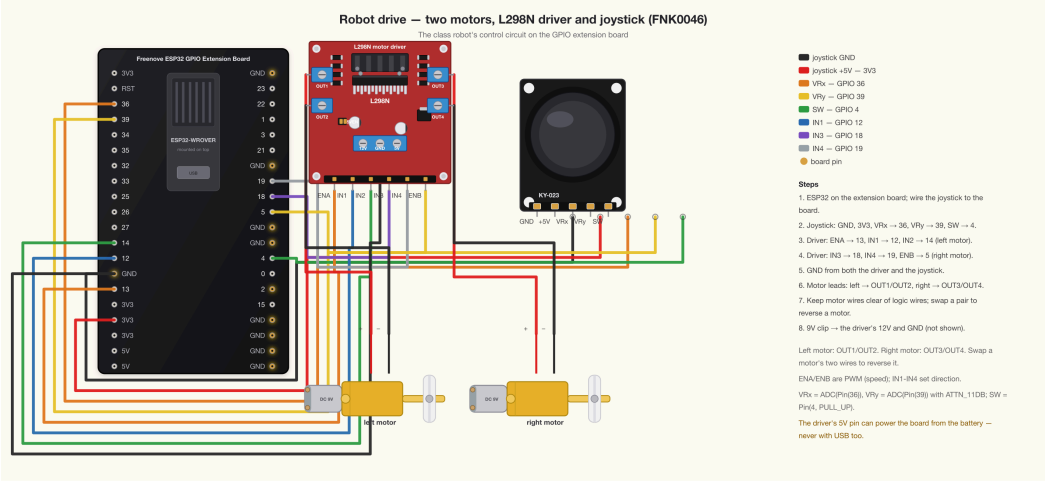
The stick drives

Now the drive train: forward and back, and pivoting turns with differential steering.

THE PROMPT

“Now drive the robot from the joystick: up and down go forward and back, left and right pivot it with differential steering. Stop when the stick is centered. I have a L298N driver to control the motors.”

This is the drive half of the robot — everything from step 1 stays wired.



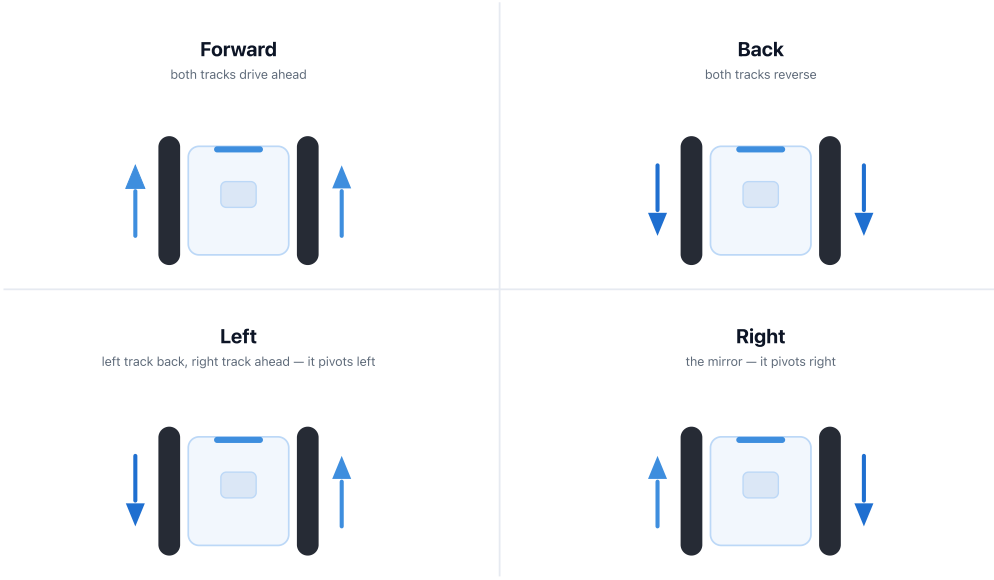
Check it before you trust it

Wheels off the ground first, then every direction on the floor.

THE PROMPT

“Add a self-test that drives forward, then backwards, then turns left and right — about a second each — and prints each step, so I can watch and check.”

Something off? Say exactly what you saw — “the left wheel spins backwards when I push up” — then re-test. One fix at a time.



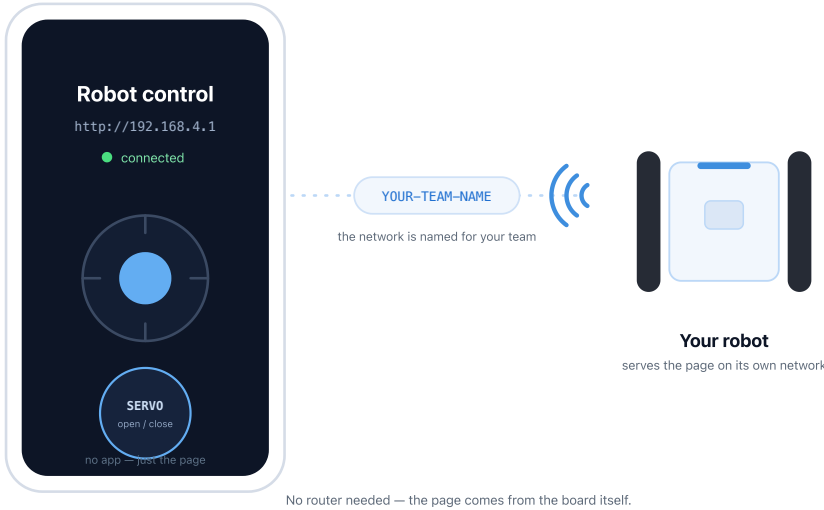
Make the robot the network

The hotspot is your team's name — your phone talks straight to the board.

THE PROMPT

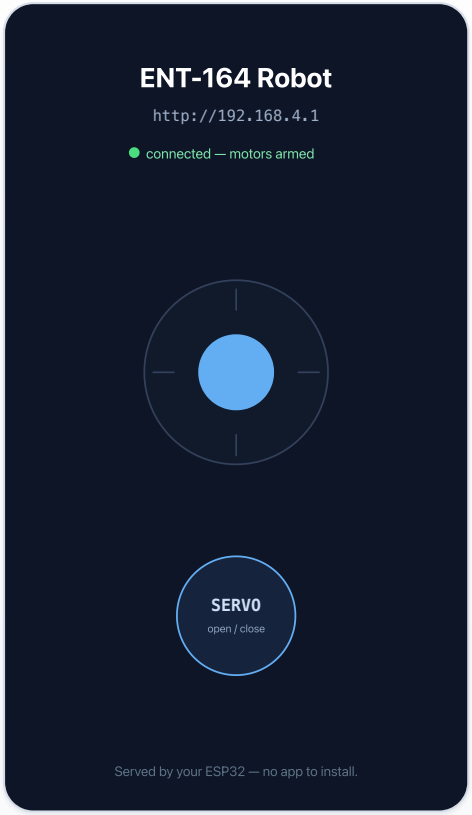
"Finally, make the robot its own network: create a WiFi hotspot on the device named [fill in your team name], and serve the joystick and button UI over it. When my phone joins the robot's WiFi, the control page opens and drives the robot — no other network needed."

Replace the highlighted placeholder with your team's name — it becomes the WiFi network everyone joins.



Every move is a request

The page has two controls: the stick drives, the button works the servo.



WHAT EACH CONTROL DOES

Drag the stick → the motors follow your thumb

Let go → everything stops

Press the button → the servo opens or closes

Every drag and press is a request the board answers itself. The whole app is one page — nothing to install.

What happened?

Every tap on your phone becomes a request the ESP32 answers.

A web server starts

1

Your ESP32 runs a web server that renders a web page. No separate computer needed.

The page gets an address

2

The board serves the page from the network it creates — the address the code prints, like `http://192.168.4.1`.

Your phone renders it

3

Your phone's browser renders the HTML that the ESP32's web server generates.

Requests round-trip

4

Interacting with the page sends a request to the ESP32 (its address is `http://192.168.4.1`). It reads the request, sets the pins, and answers with the next page state.

Other things to try

Once your device is online, the browser becomes a remote control for anything you build.

01

Visualize sensor data

A dashboard with temperature history charts.

02

Joystick driving

Swap the buttons for a virtual joystick: drag to steer, release to stop.

03

Manage settings on the phone

Save them to the device: set a thermostat temperature, or set an alarm clock.

04

Device state visualization

See the status of an LED (on/off) or the current servo angle.

Digital Making reflection

A short written reflection on the digital making so far — the electronics, coding with AI, and going wireless.

01

Catch the surprise

Find a moment when the circuit, the code, or the AI didn't do what you pictured. What did you notice, and what did you do next?

02

Think on your feet

Point to a decision you made mid-build without stopping to plan it. What were you reacting to?

03

Look back

Across the electronics, the AI sessions, and going wireless: what would you keep exactly as it is, and what would you change — and why?

HOW TO WRITE IT

Ground every answer in a specific moment — the wiring bench, the opencode chat, the first time your phone opened the page. A few honest paragraphs beat a polished essay.

DUE THU, OCT 22

Upload your reflection as a one-page document to the Canvas assignment, by noon. Reflections are part of your participation grade.

NEXT WEEK · CLASS 7

Meet directly at Nolop

No classroom first — come straight to the makerspace with everything you've made.
The whole session is the robot: assembling, testing, and fixing.

Thu, Oct 22 · 1:20–4:20pm

BEFORE YOU GO

Questions?

Then get your device online. Next week: Final Robot Assembly and Testing. Integrate everything you've learned into a working robot.

Individual assignment: electronics project. Present a working circuit.

WHERE TO FIND US

Office hours
`go.tufts.edu/meet_tvdv`

Email
`thomas.van_de_velde@tufts.edu`

Course materials
tuftsmaker.github.io/ENT-164